

Legality

Rooting a phone is allowed legally almost everywhere in the world. In fact, in some European countries, replacing the original operating system with another does not void the statutory warranty that covers the hardware of the device.

Security concerns

Rooting essentially allows applications to perform actions that require administrative privileges. The real reason for disallowing all applications from obtaining these privileges is to ensure air-tight security in this amazing operating system. Although Android has not been plagued by viruses, Trojans, spyware, malware and other members of the digital disease-causing pathogens, it is only a matter of time before anti-social elements try to exploit the innocent. Rooting your device is like giving ammunition to these elements on a silver platter.

What do you gain?

Despite these drawbacks, a number of users continue to root their phones or tabs. The gains are listed below.

Delete those pesky apps

Did your manufacturer bundle your phone with promotional applications and games that cannot be deleted? Geeks refer to these apps as bloatware, as they usually take too much space on the device's memory. So, it is always advisable to remove all that you are never likely to use. However, you have to be especially careful not to delete any system app that can affect the basic functionality of your device. What you can safely remove are games and apps that have the manufacturer's brand name.

Take essential back-ups

Rooting your device takes the performance of system and data back-ups to a completely new level. In addition to backing up user data, you can also back up applications and settings. Also, rooting allows you to take Nandroid back-ups, which is essentially a back-up of the complete operating system. The name Nandroid is a portmanteau of NAND, which is the widely used Flash memory on these devices, and Android.

Apps galore!

If you think that 1,000,000 (a million) apps and 50,000,000 downloads would be enough for open source enthusiasts, you are sorely mistaken. In addition to Google Play, there are several app distributors and stores available online. The most famous among these is F-Droid, an alternative software repository comprising only free, open source software. A quick search would reveal that most of the apps available on these stores require super-user permissions.

Playing dress-up

Almost every hardware manufacturer bundles Android devices with tones, themes and skins that are specific to their organisation. However, you are not allowed customisation

beyond a certain level. By rooting the device, not only can you change the way your system sounds or looks, you can also use the customisation options provided by other companies.

Staying up-to-date

As manufacturers come up with newer devices, it is customary for them to stop providing updates for the older ones. Also, there are several devices that possess the hardware specifications to run newer versions of the Android operating system but are stuck without any updates because of the business gimmicks of these manufacturers. Rooting allows you to install not only the new features but a tweaked kernel or a new operating system altogether.

Custom ROMs

Almost all phone and tablet manufacturers modify the Android operating system and bundle the device with a custom made OS with their own branding. The basic Android version specific to a device is known as a stock ROM. Most stock ROMs are also bundled with useful software, specific to the device's manufacturer. Custom ROMs are modified versions of the Android operating system, created for a specific device by a third-party developer. Several developers have taken great pains to make it possible for users with low-end phones to enjoy the benefits of the latest developments in the Android world. Several companies such as Cyanogenmod sprang up to cater to people facing such limitations with their own fork of the Android operating system. Although rooting is strictly not essential in order to install a stock or custom ROM, it provides you the ability to take customisation and personalisation to a whole new level.

Automation

With automation touching every aspect of life, it would be natural to also want our phones to think and act as per our wishes. There are several apps available on Google Play and other app stores that allow you to automate devices. One of the best applications for this is Tasker, which helps you create tasks for greater automation. With such apps, you can have your phone switch from data to Wi-Fi, based on location. You can combine task killer apps and ensure that your tablet consumes the least possible battery, especially when you're not using it. Rooting lifts the software restrictions and renders anything that's within the limitations of the hardware possible.

The preparations

Before you actually go ahead with rooting your device, there are several steps to be followed. The most important is to ensure that you install the latest drivers for your device on the computer to be used in this process. You must also download and install the latest Android SDK from the <https://developer.android.com/sdks/index.html> website. In addition, you need to put your phone into the USB debugging mode. To do so, enter the *Developer Options* in the system settings. You will be required to enable *Developer Options* and then enable *USB debugging*. A USB cable will also be required.